THE HOFFMAN–SINGLETON GRAPH REFUTES WOW-284

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ABSTRACT. WOW-284 asserts that the minimum dual degree of every connected graph of order at least three and girth at least five is at most the negative of its least distance eigenvalue. We record a counterexample on 50 vertices. The graph is given by an explicit coordinate construction over $\mathbb{F}_5$. It is 7-regular, has girth five, and has distance spectrum

$$\{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}.$$

Consequently its minimum dual degree is 7, whereas the negative of its least distance eigenvalue is 4. The strict gap is therefore 3. The graph is the Hoffman–Singleton graph, but the proof uses no classification or named-graph property: a direct common-neighbor certificate yields connectedness, girth, and the full spectrum. Exact verification code builds the distance matrix independently by breadth-first search and checks the characteristic polynomial over the integers.

INTRODUCTION

Let G be a finite simple connected graph. Write $D(G)$ for its distance matrix and order its distance eigenvalues as

$$\partial_1(G) \geq \partial_2(G) \geq \cdots \geq \partial_n(G).$$

For a vertex v, its *dual degree* is the average degree of its neighbors,

$$d^*(v) = \frac{1}{d(v)} \sum_{u \in N(v)} d(u), \qquad \delta^*(G) = \min_{v \in V(G)} d^*(v).$$

Aouchiche and Hansen record the following Graffiti conjecture in their survey of distance spectra [1, Conjecture 7.16]; they attribute it to Fajtlowicz's 1998 *Written on the Wall* report [2]. The associated public catalogue is DeLaViña's *Written on the Wall II* site [3].

Conjecture 0.1 (WOW-284)**.** *If G is connected, has $n \geq 3$ vertices, and has girth $g(G) \geq 5$, then*

$$\delta^*(G) \leq -\partial_n(G).$$

The same survey states that the conjecture was open there and notes equality for the Petersen graph. The degree-7 Moore graph constructed by Hoffman and Singleton [4] gives a strict violation.

There is an important prior spectral calculation. Howlader and Panigrahi explicitly give the distance spectrum $\{91, (-4)^{(28)}, 1^{(21)}\}$ of the Hoffman–Singleton graph [5, Theorem 2.5(1b)]. Combining that calculation with 7-regularity already disproves Conjecture 0.1. The present note records this connection and supplies a self-contained coordinate certificate, so neither the graph identification nor the earlier spectral formula is needed as a premise. No claim is made that the spectrum is new, that this observation has priority over every unpublished observation, or that 50 is the smallest possible order.

Theorem A. *There is a simple connected graph G on 50 vertices such that*

$$g(G) = 5, \qquad \delta^*(G) = 7, \qquad \partial_{50}(G) = -4.$$

In particular, G violates WOW-284 by the strict gap $\delta^*(G) + \partial_{50}(G) = 3$.

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1. Coordinate construction

All subscripts in this section lie in the field $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. Introduce vertices

$$V(G) = \{P_{i,j} : i, j \in \mathbb{F}_5\} \,\dot\cup\, \{Q_{k,\ell} : k, \ell \in \mathbb{F}_5\}.$$

Thus $|V(G)| = 50$. The edges are the following unordered pairs:

$$E_P = \big\{\{P_{i,j}, P_{i,j+1}\} : i, j \in \mathbb{F}_5\big\},$$
$$E_Q = \big\{\{Q_{k,\ell}, Q_{k,\ell+2}\} : k, \ell \in \mathbb{F}_5\big\},$$
$$E_X = \big\{\{P_{i,j}, Q_{k,ik+j}\} : i, j, k \in \mathbb{F}_5\big\}.$$

Set $E(G) = E_P \cup E_Q \cup E_X$. For a fixed numerical labeling, with representatives in $\{0, 1, 2, 3, 4\}$, use

$$P_{i,j} \longleftrightarrow 5i + j, \qquad Q_{k,\ell} \longleftrightarrow 25 + 5k + \ell.$$

The complete neighborhood formulas are

$$N(P_{i,j}) = \{P_{i,j-1}, P_{i,j+1}\} \cup \{Q_{k,ik+j} : k \in \mathbb{F}_5\}, \tag{1}$$
$$N(Q_{k,\ell}) = \{Q_{k,\ell-2}, Q_{k,\ell+2}\} \cup \{P_{i,\ell-ik} : i \in \mathbb{F}_5\}. \tag{2}$$

Lemma 1.1. *The construction defines a simple 7-regular graph with 175 edges.*

Proof. Every listed edge has distinct endpoints. Within each of the five fixed P-layers, addition by 1 gives a 5-cycle; within each fixed Q-layer, addition by 2 gives a 5-cycle. The cross edges have one endpoint of each type. The set notation removes any possibility of parallel edges.

Equations (1)–(2) list seven distinct neighbors at every vertex. There are 25 edges of type E_P, 25 of type E_Q, and 125 cross edges, hence 175 in total. Equivalently, the handshake identity gives $50 \cdot 7/2 = 175$. □

2. The common-neighbor certificate

Proposition 2.1. *For any two distinct vertices x, y,*

$$|N(x) \cap N(y)| = \begin{cases} 0, & x \sim y, \\ 1, & x \nsim y. \end{cases}$$

Proof. First consider two P-vertices $P_{i,j}$ and $P_{i',j'}$. If $i = i'$, their layer is a 5-cycle. Adjacent pairs have no common P-neighbor, while nonadjacent pairs have exactly one. They cannot have a common Q-neighbor: for fixed i and k, the equation $Q_{k,\ell} \sim P_{i,j}$ determines $j = \ell - ik$ uniquely. If $i \neq i'$, there is no common P-neighbor, and a common Q-neighbor must satisfy

$$ik + j = i'k + j'.$$

This has the unique solution

$$k = \frac{j' - j}{i - i'}, \qquad \ell = ik + j,$$

because $\mathbb{F}_5$ is a field.

The argument for two Q-vertices is dual. If their first coordinates agree, they lie in the 5-cycle generated by steps ± 2. Adjacent pairs have no common Q-neighbor and nonadjacent pairs have exactly one; no common P-neighbor is possible. If $k \neq k'$, a common $P_{i,j}$ is determined uniquely by

$$\ell = ik + j, \qquad \ell' = ik' + j,$$

namely

$$i = \frac{\ell - \ell'}{k - k'}, \qquad j = \ell - ik.$$

It remains to compare $P_{i,j}$ with $Q_{k,\ell}$. Put

$$r = \ell - (ik + j) \in \mathbb{F}_5.$$

The pair is adjacent precisely when $r = 0$. A common P-vertex must be $P_{i,j+s}$ with $s \in \{\pm 1\}$, and it is adjacent to $Q_{k,\ell}$ precisely when $r = s$. Hence there is exactly one common P-neighbor if $r \in \{\pm 1\}$, and none otherwise. A common Q-vertex must be $Q_{k,\ell+t}$ with $t \in \{\pm 2\}$, and it is adjacent to $P_{i,j}$ precisely when $t = -r$. Hence there is exactly one common Q-neighbor if $r \in \{\pm 2\}$, and none otherwise. Finally,

$$\mathbb{F}_5 = \{0\} \,\dot\cup\, \{\pm 1\} \,\dot\cup\, \{\pm 2\},$$

which proves every cross case and completes the certificate. □

Corollary 2.2. *The graph is connected, has diameter two, and has girth five.*

Proof. Every pair of distinct vertices is adjacent or has a common neighbor, so the graph is connected and has diameter at most two. It is not complete, since it is 7-regular on 50 vertices, so its diameter is exactly two.

A triangle would give an adjacent pair a common neighbor. In a 4-cycle, two opposite vertices would have the other two cycle vertices as distinct common neighbors. Both possibilities contradict Proposition 2.1. Thus $g(G) \geq 5$. On the other hand, for every fixed i,

$$P_{i,0}P_{i,1}P_{i,2}P_{i,3}P_{i,4}P_{i,0}$$

is a 5-cycle, so $g(G) = 5$. □

3. Exact spectral calculation

Let A be the adjacency matrix, let I be the 50×50 identity matrix, and let J be the all-ones matrix. Proposition 2.1 gives, entry by entry,

$$A^2 = 6I - A + J. \tag{3}$$

Indeed, the diagonal entries on both sides are 7, the entries indexed by edges are 0, and all other off-diagonal entries are 1.

Since the graph is connected and 7-regular, $A\mathbf{1} = 7\mathbf{1}$, with this eigenvalue simple. On $\mathbf{1}^\perp$, equation (3) becomes

$$A^2 + A - 6I = 0.$$

Thus the other adjacency eigenvalues are 2 and -3. If their multiplicities are m_2 and m_{-3}, then

$$m_2 + m_{-3} = 49, \qquad 7 + 2m_2 - 3m_{-3} = \operatorname{tr} A = 0.$$

Consequently

$$\operatorname{Spec}(A) = \{7^{(1)}, 2^{(28)}, (-3)^{(21)}\}. \tag{4}$$

By Corollary 2.2, two distinct vertices are at distance one on edges and distance two on nonedges. Therefore

$$D = A + 2(J - I - A) = 2J - 2I - A. \tag{5}$$

On $\mathbf{1}$, the distance matrix has eigenvalue

$$2 \cdot 50 - 2 - 7 = 91.$$

If $x \in \mathbf{1}^\perp$ and $Ax = \theta x$, then

$$Dx = (-2I - A)x = (-2 - \theta)x.$$

Using (4), we obtain

$$\operatorname{Spec}(D) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}, \tag{6}$$

or equivalently

$$\det(xI - D) = (x - 91)(x - 1)^{21}(x + 4)^{28}.$$

Proof of Theorem A. Corollary 2.2 gives connectedness and $g(G) = 5$. By Lemma 1.1, every vertex and every one of its neighbors has degree 7, so

$$d^*(v) = \frac{1}{7} \sum_{u \in N(v)} 7 = 7$$

for every v, and hence $\delta^*(G) = 7$. Equation (6) gives $\partial_{50}(G) = -4$. Thus

$$\delta^*(G) = 7 > 4 = -\partial_{50}(G), \qquad \delta^*(G) + \partial_{50}(G) = 3 > 0.$$

This is a strict counterexample to WOW-284. □

As a further exact certificate, the eigenvalues of $D + 7I$ are 98, 8 with multiplicity 21, and 3 with multiplicity 28. Thus $D + 7I$ is positive definite.

4. Computational certificate

The proof above is entirely analytic. The accompanying Python code supplies a separate exact check from the coordinate rules. It performs the following steps:

(1) constructs the 175 unordered edges and checks all 50 degrees;
(2) checks the common-neighbor count for every unordered vertex pair;
(3) constructs all 2,500 distance entries by integer breadth-first search;
(4) computes the distance characteristic polynomial symbolically; and
(5) computes an exact rational LDL^{T} decomposition of $D + 7I$ and checks that all 50 pivots are positive.

The symbolic steps use SymPy [6]. The expected output is

$$(x - 91)(x - 1)^{21}(x + 4)^{28}$$

together with 50 positive rational pivots. Because the distance matrix is built by BFS, this check does not use equation (5) or the spectral derivation above. Numerical agreement is not treated as a proof; all calculations in the certificate are exact.

Scope, provenance, and disclosures

The counterexample has order 50. No minimality claim is made, and no exhaustive search over smaller graphs is part of this note. The repository's source ledger distinguishes the published distance spectrum from the explicit WOW-284 connection and records the limits of the targeted priority search.

OpenAI ChatGPT-5.6 Sol Pro assisted with adversarial proof checking and finding the exact graph construction. No AI system is an author. Samuil Petkov is the sole named author and assumes full responsibility for the mathematics, citations, attribution, and conclusions.

The author received no funding for this work and declares no competing interests. The manuscript source, exact verification code, machine-readable certificates, and build instructions are available in the public repository `github.com/SamPetkov/wow284`.

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